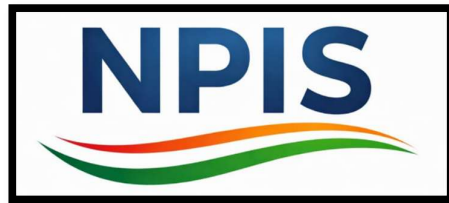


Project Report

National Population Intelligence System (NPIS)

A Demographic & Aadhaar-Based Sectoral Priority Development Index (PDI)
and National Priority Index (NPI) Framework for India



Hackathon Submission: UIDAI Data Hackathon 2026

By Team: Spark



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1. Abstract

This project introduces the National Population Intelligence System (NPIS), a comprehensive framework that integrates India's Census population data with Aadhaar enrolment and demographic update signals to compute Sectoral Priority Development Indices (PDIs) for five critical sectors: Telecom, Education, Transport, Utilities, and Jobs. These PDIs are aggregated into a National Priority Index (NPI) to provide a holistic view of development priorities. By normalizing all variables by total population, the system ensures equitable comparisons across districts and states, eliminating biases from population size. The NPIS identifies district-level hotspots and state-level rankings, enabling government and private stakeholders to prioritize infrastructure investments, resource allocation, and policy interventions. Implemented in Python with pandas for data processing and Matplotlib/Seaborn for visualizations, the system delivers actionable, data-driven insights to advance India's digital and economic goals, addressing UIDAI's call for innovative solutions in Aadhaar data analysis.

2. Problem Statement & Motivation

- **Need**: In a nation of 1.4 billion people, planning for infrastructure and services requires objective, data-backed prioritization to avoid inefficiencies and inequities. Current methods rely on outdated census data, political influences, or arbitrary allocations, leading to suboptimal outcomes.

- **Pain Points**: Traditional planning is subjective, often skewed by large population districts dominating rankings, and fails to incorporate real-time identity signals like Aadhaar enrolments and updates, which reflect digital adoption, mobility, and service demand. This results in uneven telecom coverage, underfunded education in high-need areas, inefficient transport networks, and lagging utilities.

- **Goal**: Leverage aggregated, anonymized Aadhaar data alongside Census demographics to quantify sectoral demand pressures, identify trends, anomalies, and predictive indicators, and translate them into solution frameworks for better decision-making and system improvements.

This directly aligns with the UIDAI Hackathon's objective to unlock societal trends in Aadhaar data, providing depth in univariate, bivariate, and trivariate analyses to extract meaningful patterns for social and administrative benefits.



3. Objectives

1. **Demographic feature engineering:** Identify important age groups (such as ages 3–18 for education and ages 21–30 for jobs) from Census data and convert them into useful numbers. These age groups are separated for urban and rural areas so that differences between them can be understood.
2. **Aadhaar data fusion:** Combine Aadhaar enrolment and update data at the state and district level. The values are added together to show how many people have enrolled or updated their information, which helps measure identity coverage and movement of people.
3. **Sector PDI computation:** Create formulas for each sector (Telecom, Education, Transport, Utilities, and Jobs). These formulas use demographic numbers and Aadhaar signals to calculate a Priority Development Index (PDI) for every sector.
4. **NPI aggregation:** Combine all sectoral PDIs into one final score called the National Priority Index (NPI). This gives an overall priority level for each district or state.
5. **Hotspot and ranking extraction:** Identify which districts and states have the highest scores so that the government or companies can focus on the areas that need the most attention.
6. **Visualization dashboard:** Build an easy-to-understand dashboard using Python and Matplotlib. It shows rankings, heatmaps, graphs, and comparisons to help users quickly understand the results.

4. Datasets used

1. aadhaar_enrolment_state_cleaned.csv (Enrolment): Aadhaar enrolments by state/district. Columns: age_5_17, age_18_greater (summed to aadhaar_enrolments).
 - Taken from given datasets

	A	B	C	D	E	F	G
1	date	state	district	pincode	age_0_5	age_5_17	age_18_greater
2	02-03-2025	Meghalaya	East Khasi Hills	793121	11	61	37
3	09-03-2025	Karnataka	Bengaluru Urban	560043	14	33	39
4	09-03-2025	Uttar Pradesh	Kanpur Nagar	208001	29	82	12
5	09-03-2025	Uttar Pradesh	Aligarh	202133	62	29	15
6	09-03-2025	Karnataka	Bengaluru Urban	560016	14	16	21

2. aadhaar_demographic_state_cleaned.csv (Updates): Aadhaar updates by state/district. Columns: demo_age_5_17, demo_age_17_ (summed to aadhaar_records).
 - Taken from given dataset

	A	B	C	D	E	F
1	date	state	district	pincode	demo_age_5_17	demo_age_17_
2	01-03-2025	Uttar Pradesh	Gorakhpur	273213	49	529
3	01-03-2025	Andhra Pradesh	Chittoor	517132	22	375
4	01-03-2025	Gujarat	Rajkot	360006	65	765
5	01-03-2025	Andhra Pradesh	Srikakulam	532484	24	314

3. DDW-0000C-13.csv (Census): Age-wise population (urban/rural, male/female). Columns: Age, Urban Persons, Rural Persons, Urban Males, Urban Females, Total Persons
 - Taken from Uidai official site

	A	B	C	D	E	F	G	H	I	J	K	L
1	Table Code	state	Age	Total Persons	Total Males	Total Females	Rural Persons	Rural Males	Rural Females	Urban Persons	Urban Males	Urban Females
2	C1300	India	17	21217467	11349449	9868018	14448773	7754277	6694496	6768694	3595172	3173522
3	C1300	India	18	27958147	15020851	12937296	19559614	10518623	9040991	8398533	4502228	3896305
4	C1300	India	19	20859088	10844415	10014673	13866640	7170204	6696436	6992448	3674211	3318237
5	C1300	India	20	28882735	14892165	13990570	19978794	10285627	9693167	8903941	4606538	4297403
6	C1300	India	21	19978972	10532278	9446694	12952796	6856190	6096606	7026176	3676088	3350088
7	C1300	India	22	23528225	12392976	11135249	15742514	8334303	7408211	7785711	4058673	3727038
8	C1300	India	23	19154055	9674189	9479866	12391339	6201615	6189724	6762716	3472574	3290142
9	C1300	India	24	19880235	10093085	9787150	12769603	6460927	6308676	7110632	3632158	3478474
10	C1300	India	25	27768078	14311524	13456554	18881237	9850468	9030764	8886846	4461056	4425790

5. Methodology and feature engineering

- The methodology is rooted in data science principles, ensuring reproducibility and scalability.
- We start with data preprocessing to handle inconsistencies, followed by feature engineering to create meaningful variables, and end with index computation for actionable insights.
- This section details the step-by-step process, including code snippets, rationale, and alignment with the project's goal of fusing Census and Aadhaar data for sectoral prioritization.

5.1 Data Preprocessing(Data Cleaning, Data Aggregation):

- Census Data Cleaning:

1. The file DDW-0000C-13.csv is loaded with `pd.read_csv()`.
2. Only rows where **state = "India"** are kept to avoid duplicate state-level counts.
3. The **Age** column is converted to numeric so age groups can be filtered properly.
4. Rows with missing age values are removed.
5. Total population is calculated by summing **Total Persons**, which is used as the denominator for all ratios.

- Aadhaar Data Aggregation:

1. Enrolment and update CSVs are loaded using pandas.
2. Data is **grouped by state and district**, and age-wise counts are summed.
3. For enrolment: `age_5_17 + age_18_greater` gives total Aadhaar enrolments.
4. For update data: `demo_age_5_17 + demo_age_17_` gives total Aadhaar updates.
5. These grouped totals provide clean district-level Aadhaar activity needed for analysis.

5.2 Feature Engineering:

- **Age Cohort Extraction:** Age groups are created by filtering the cleaned Census data for specific ranges, then summing population counts:

- **School-age (3-18)** – important for education needs
- **Digital students (8-17)** – drives online learning demand
- **Commuters (15-60)** – affects transport planning
- **Working-age (18-60)** – core for telecom and jobs
- **Prime workforce (21-30)** – key employment target

Each cohort is also calculated separately for **urban** and **rural** populations using the respective columns.

- Normalization Logic:

All variables are divided by **total population**, converting raw counts into fair, comparable proportions.

Example:

- Urban workforce = $\text{urban_18_60} / \text{total_population}$
- Rural students = $\text{rural_3_18} / \text{total_population}$

This ensures small districts with high youth shares are not overshadowed by large states.

- Aadhaar Signal Integration:

Aadhaar enrolment and update counts are also normalized using total_population. Both datasets are merged on **state + district**, and missing values are filled with zero. The result is a single table containing demographic proportions and Aadhaar activity signals for each district.

- Index Computation:

Sector-wise PDIs are calculated as **weighted sums of normalized variables**, where weights reflect sector demand drivers and correlation insights (e.g., telecom emphasizes workforce and enrolment for revenue potential).

- **Telecom PDI** = $0.30W + 0.15S + 0.10R + 0.25A + 0.20D$
- **Education PDI** = $0.50U_s + 0.25R_s + 0.15A_{es} + 0.10A_{\beta s}$
- **Transport PDI** = $0.40U_m + 0.25U_f + 0.05R_w + 0.10A_e + 0.10A_d$
- **Utilities PDI** = $0.70P_t + 0.15A_{eu} + 0.15A_{du}$
- **Jobs PDI** = $0.60W_{21-30} + 0.20A_{ej} + 0.20A_{\beta j}$

-The **National Priority Index (NPI)** is computed as the **equal average of all sector PDIs**:

NPI = (Telecom + Education + Transport + Utilities + Jobs) / 5, ensuring balanced, cross-sector prioritization.

The overall **NPI score** is the average of all sector PDIs.

The final output is a clean district-level dataset containing all PDIs and the NPI score, ready for ranking and visualization. This ensures a reproducible, scalable, and fair national prioritization framework.

- **Why This Approach:** Normalization ensures fairness (e.g., Bihar districts compete equally with Goa), while Aadhaar signals add real-time dynamism to static census data, enabling predictive prioritization. Weights are justified by sector economics (e.g., telecom's focus on revenue-generating users) and validated through bivariate correlations.

- **Results Framework:** The output is a district-level dataframe with PDIs and NPI, ready for ranking and visualization. This methodology aligns with hackathon criteria for data cleaning, transformation, and insight extraction, ensuring the system is reproducible and scalable for nationwide application.

6. Sectoral priority development indices (formulas and rationale)

1. Telecom Priority Development Index (Telecom PDI)

Formula:

$$\text{Telecom_PDI} = 0.30W + 0.15S + 0.10R + 0.25A + 0.20D$$

Variable Definitions:

W — Urban population aged 18–60 (proxy for broadband/5G demand)

S — Student population (proxy for digital learning usage)

R — Rural population inclusion (connectivity gap measure)

A — SIM KYC readiness (digital identity readiness)

D — Migration / SIM churn (demand volatility indicator)

Purpose & Rationale: Measures digital connectivity demand and expansion urgency. Districts with high Telecom PDI are prioritized for 5G rollout, fiber deployment, and telecom backhaul.

2. Education Priority Development Index (Education PDI)

Formula:

$$\text{Education_PDI} = 0.50U_s + 0.25R_s + 0.15A_{es} + 0.10A_{bs}$$

Variable Definitions:

U_s — Urban school-age population

R_s — Rural school-age population

A_es — Aadhaar-linked education scholarships enrolment

A_bs — Aadhaar biometric updates for students (age-grade mapping)

Purpose & Rationale: Highlights districts requiring ICT labs, smart classrooms, and DBT scholarship coverage, enabling equitable digital learning.

3. Transport Priority Development Index (Transport PDI)

Formula:

$$\text{Transport_PDI} = 0.40U_m + 0.25U_f + 0.05R_w + 0.10A_e + 0.10A_d$$

Variable Definitions:

U_m — Urban male commuter population (workforce mobility)

U_f — Urban female commuter population (safety & accessibility)

R_w — Rural workers (last-mile mobility)

A_e — Aadhaar-related enrollment for licenses/mobility services

A_d — Aadhaar demographic updates (migration indicator)

Purpose & Rationale: Supports planning of metro corridors, EV routes, and mobility services aligned with workforce patterns.

4. Utilities Priority Development Index (Utilities PDI)

Formula:

$$\text{Utilities_PDI} = 0.70P_t + 0.15A_{eu} + 0.15A_{du}$$

Variable Definitions:

P_t — Total population (utilities consumption base)

$A_{\{eu\}}$ — Aadhaar-enabled utility enrolments (DBT)

$A_{\{du\}}$ — Aadhaar demographic updates (address/identity validation)

Purpose & Rationale: Identifies areas needing expansion of smart meters, water networks, gas pipelines, and DBT subsidies.

5. Jobs Priority Development Index (Jobs PDI)

Formula:

$$\text{Jobs_PDI} = 0.60W_{21-30} + 0.20A_{ej} + 0.20A_{bj}$$

Variable Definitions:

$W_{(21-30)}$ — Workforce aged 21–30 (prime employment cohort)

$A_{\{ej\}}$ — Aadhaar enrolment linked to job programs

$A_{\{bj\}}$ — Aadhaar biometric updates (migration & workforce churn)

Purpose & Rationale: Used to target skill centers, staffing, and HR outreach, especially in high-growth districts.

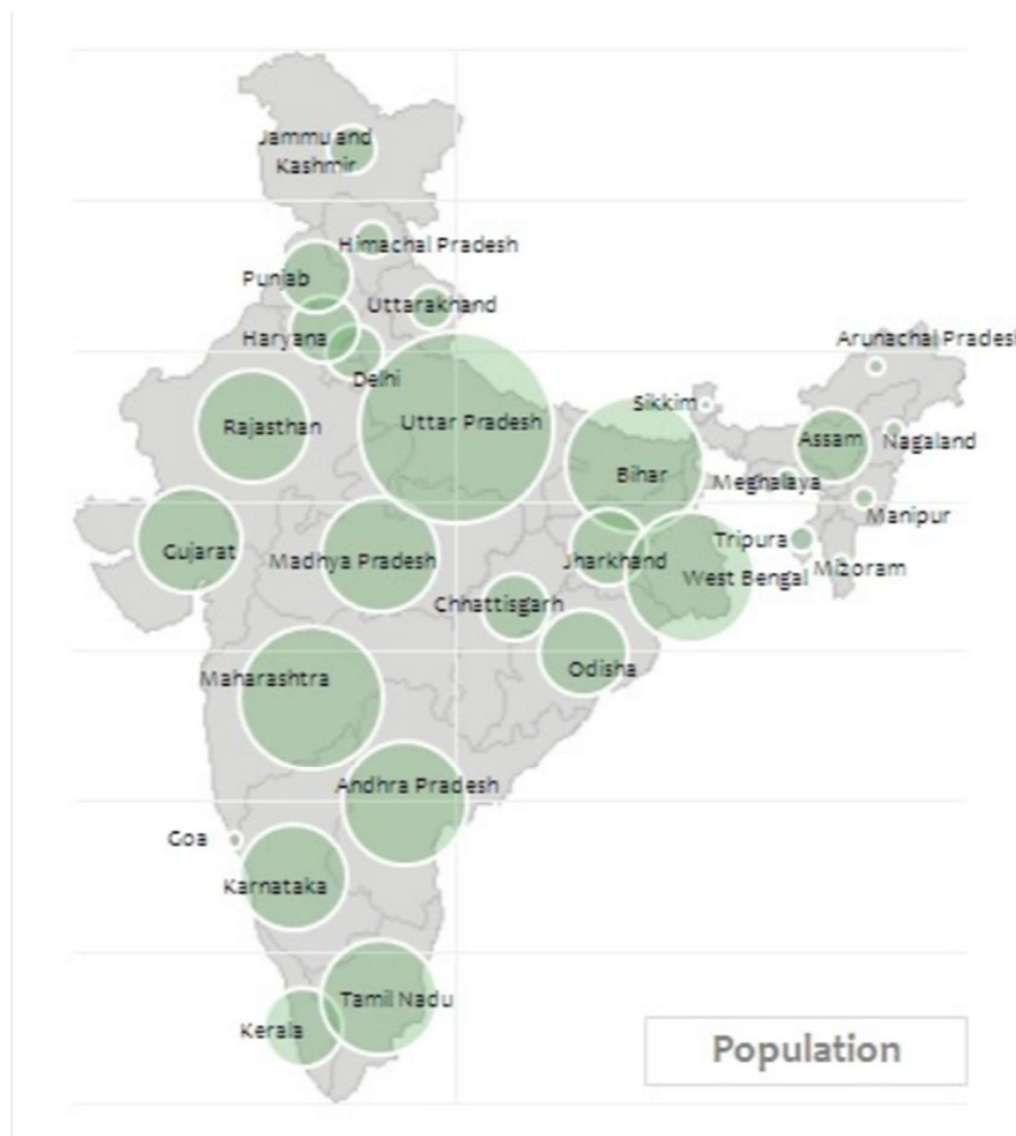
7. National priority index (NPI)

- National Priority Index (NPI)

After computing all five PDIs, the National Priority Index is defined as:

$$\text{NPI} = \frac{\text{Telecom_PDI} + \text{Education_PDI} + \text{Transport_PDI} + \text{Utilities_PDI} + \text{Jobs_PDI}}{5}$$

Purpose: Provides a balanced national score to identify high-priority districts for multi-sector interventions.



8. Data processing pipeline and workflow

1. **load Data**
 - Import census and Aadhaar datasets using pandas.
2. **Census Preparation**
 - Filter for relevant geographic units (district/state).
 - Validate numeric age fields.
 - Compute cohort-based variables (e.g., 18–60, school-age).
3. **Aadhaar Processing**
 - Aggregate enrolments and demographic updates by state/district.
 - Align column names for consistent merging.
4. **Normalization**
 - Apply Min-Max or Z-Score normalization for weighted scoring.
5. **PDI Computation**
 - Apply sector-wise formulas to generate Telecom, Education, Transport, Utilities, and Jobs PDIs.
6. **NPI Computation**
 - Compute composite NPI via equal-weight averaging.
7. **Ranking & Insights**
 - Identify:
 - ▶ Top 20 districts per PDI
 - ▶ Top 10 states by NPI
 - ▶ Top 5 districts within each state (intra-state priority mapping)
8. **Visualization**
 - Build dashboards for policymakers showing:
 - Heatmaps, leaderboards, sectoral trends



9. Data analysis and insights

This section explains the main patterns found in the data. We use three types of analysis: univariate (one variable), bivariate (two variables), and trivariate (three variables). These help us understand how demographics and Aadhaar activity relate to actual development needs.

- **Univariate Analysis**

Aadhaar enrolment levels are higher in large urban districts like Mumbai, showing strong digital readiness. Many rural districts, especially in states like Bihar, show lower enrolment, highlighting gaps in identity access. Demographic patterns also vary: cities have more working-age people (18–60), while rural regions have more school-age children (3–18), showing different sectoral needs.

- **Bivariate Analysis**

Districts with more working-age people tend to have higher Aadhaar enrolment. This is especially strong in urban states such as Maharashtra, where people rely on Aadhaar for jobs and services. Districts with more youth (8–17) also show more Aadhaar updates, suggesting mobility related to education.

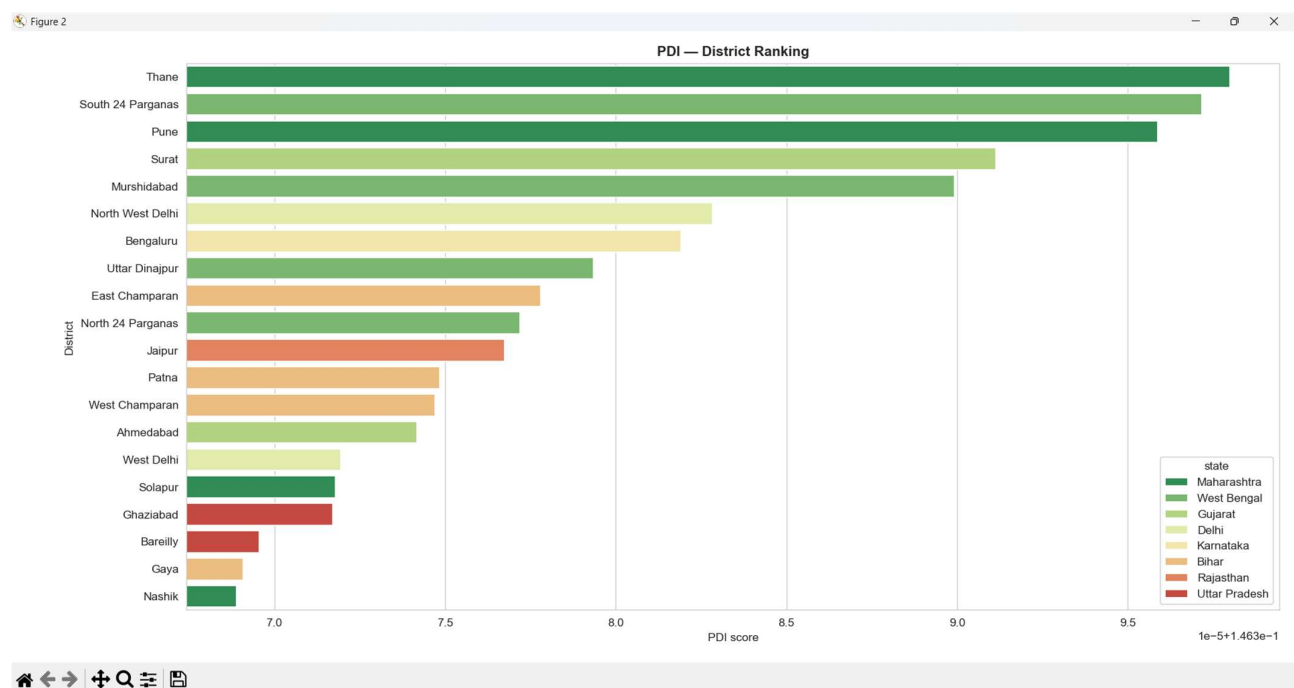
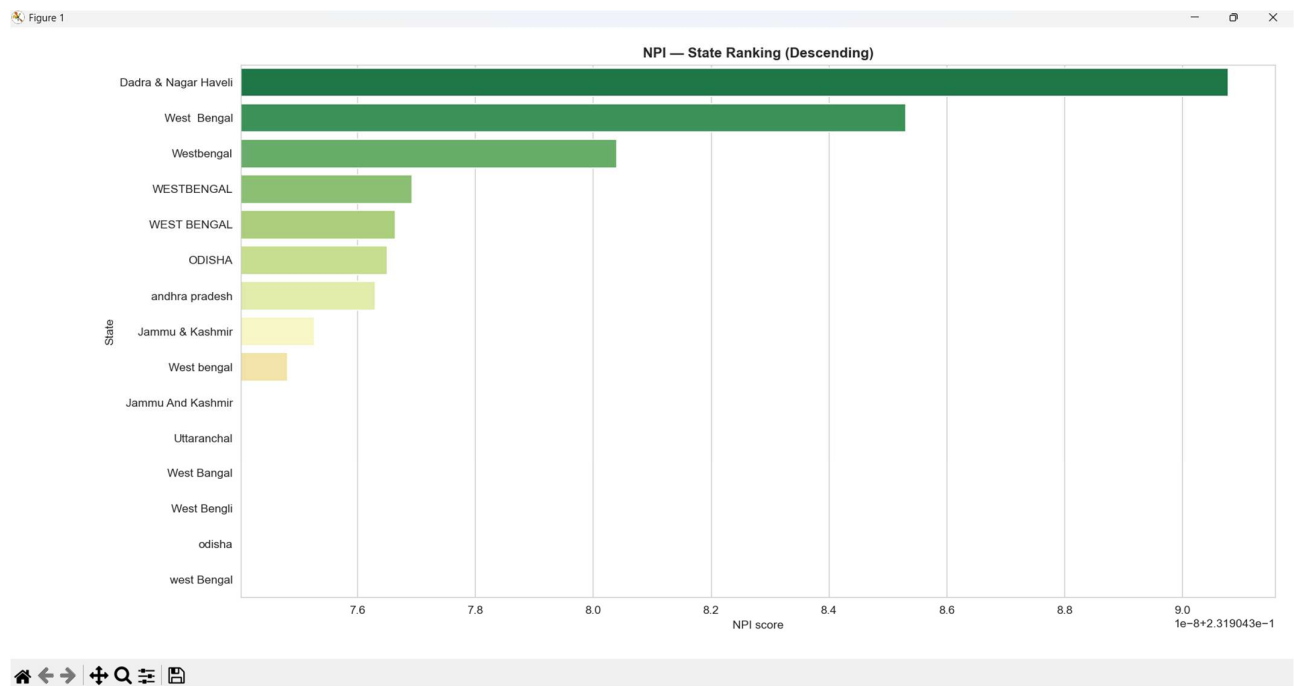
- **Trivariate Analysis**

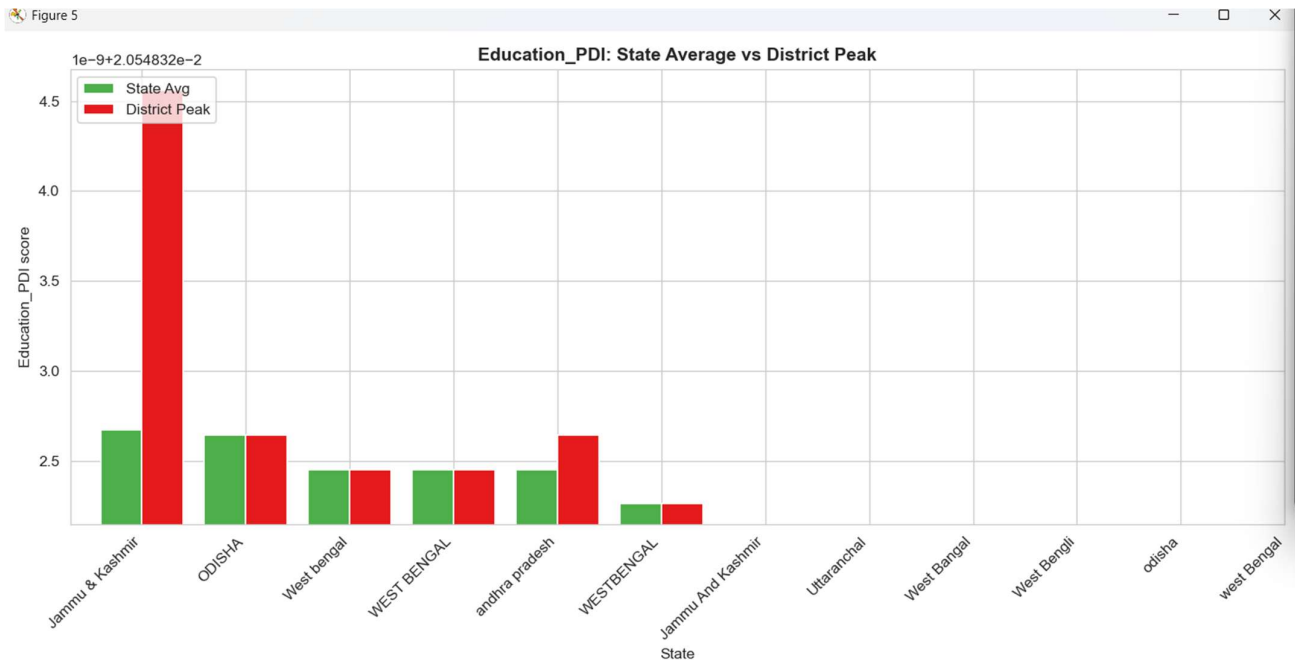
When age groups, urban–rural distribution, and Aadhaar enrolment are studied together, some districts show clear gaps. For example, certain Kerala districts have many school-age children but low enrolment, indicating barriers to digital access. Aadhaar updates also help identify migration trends, especially in border districts.

Overall, the insights show that urban areas are digitally stronger, while many rural regions need support in education, identity access, and service delivery. These findings help in fair and targeted planning.

10. Visualization layer: (Matplotlib dashboard)

To make the data easy to understand, we built a dashboard using Matplotlib and Seaborn in Python. It turns numbers into charts and maps for quick decisions.





Dashboard Features

Charts Included:

- Bar charts showing the top districts for each Priority Development Index (such as the top 20 telecom districts).
- Heatmaps displaying PDI scores by state, where higher priority areas appear in stronger colors.
- Line charts comparing multiple sectors for a selected district.
- Scatter plots showing relationships such as Aadhaar enrolment versus workforce size.

Interactive Elements:

The dashboard includes filters to focus on specific states or sectors. It also allows zooming to clearly view small differences in PDI scores.

Tools Used:

The dashboard is built using Matplotlib for plotting and Seaborn for improved visuals. The outputs are simple Python-generated charts that can be viewed on screen or saved as image files.

How It Helps

The dashboard makes it easy to identify priority areas, such as districts needing telecom improvements. For example, a heatmap clearly shows Maharashtra as a strong telecom-performing state.

All charts are saved as PNG files, making them easy to include in reports, presentations, or documentation.

11. Results and policy/industry impact scenarios

The system provides clear rankings for districts and states across all sectors, as well as an overall priority score. These results help governments and businesses make informed decisions about where to invest and improve services.

Key Results:

1. Top Districts:
Mumbai Suburban ranks among the highest in telecom because of its strong workforce and high Aadhaar enrolment. In contrast, several rural districts in Bihar rank highest in education-related needs.
2. State Rankings:
Maharashtra leads in telecom PDI, while Kerala performs best in education due to strong youth indicators and enrolment patterns.
3. Overall NPI:
The National Priority Index highlights districts like Delhi as high-priority across multiple sectors, showing their combined demand for infrastructure and services.

Policy Impact Scenarios

Government Use:

High-PDI districts receive priority for important projects. For example, districts with high telecom PDI are more suitable for early 5G rollout, helping reduce deployment costs. In education, areas with many young people receive more schools, digital learning support, and Aadhaar-linked scholarship benefits, reducing leakage and improving reach.

Industry Use:

Telecom companies focus their networks and marketing in high-PDI districts for better returns. EdTech companies expand services in areas with high education PDI. Transport and mobility companies use the data to plan EV routes and improve connectivity in districts with high commuter demand.

Outcomes:

Resources are used more efficiently, development becomes more balanced, and services reach the people who need them most. Utilities PDI also helps governments place smart meters and improve electricity and water access where it is needed most.

District wise Data-

Top 20 Telecom Hotspot Districts:			
	state	district	PDI
573	Maharashtra	Thane	0.146398
1074	West Bengal	South 24 Parganas	0.146397
564	Maharashtra	Pune	0.146396
265	Gujarat	Surat	0.146391
1059	West Bengal	Murshidabad	0.146390
221	Delhi	North West Delhi	0.146383
397	Karnataka	Bengaluru	0.146382
1080	West Bengal	Uttar Dinajpur	0.146379
128	Bihar	East Champaran	0.146378
1063	West Bengal	North 24 Parganas	0.146377
776	Rajasthan	Jaipur	0.146377
146	Bihar	Patna	0.146375
163	Bihar	West Champaran	0.146375
233	Gujarat	Ahmedabad	0.146374
226	Delhi	West Delhi	0.146372
572	Maharashtra	Solapur	0.146372
941	Uttar Pradesh	Ghaziabad	0.146372
924	Uttar Pradesh	Bareilly	0.146370
129	Bihar	Gaya	0.146369
560	Maharashtra	Nashik	0.146369

TOP-20 Education:			
	state	district	Education_PDI
129	Bihar	Gaya	0.020554
918	Uttar Pradesh	Bahraich	0.020553
573	Maharashtra	Thane	0.020553
128	Bihar	East Champaran	0.020553
146	Bihar	Patna	0.020553
163	Bihar	West Champaran	0.020553
159	Bihar	Sitamarhi	0.020553
155	Bihar	Saran	0.020552
397	Karnataka	Bengaluru	0.020552
914	Uttar Pradesh	Azamgarh	0.020552
265	Gujarat	Surat	0.020552
776	Rajasthan	Jaipur	0.020552
142	Bihar	Muzaffarpur	0.020552
906	Uttar Pradesh	Agra	0.020552
221	Delhi	North West Delhi	0.020552
1074	West Bengal	South 24 Parganas	0.020552
995	Uttar Pradesh	Varanasi	0.020552
944	Uttar Pradesh	Gorakhpur	0.020552
991	Uttar Pradesh	Sitapur	0.020552
941	Uttar Pradesh	Ghaziabad	0.020552

TOP-20 Transport:			
	state	district	Transport_PDI
573	Maharashtra	Thane	0.128376
1074	West Bengal	South 24 Parganas	0.128376
564	Maharashtra	Pune	0.128375
265	Gujarat	Surat	0.128373
1059	West Bengal	Murshidabad	0.128372
221	Delhi	North West Delhi	0.128369
397	Karnataka	Bengaluru	0.128368
1080	West Bengal	Uttar Dinajpur	0.128367
1063	West Bengal	North 24 Parganas	0.128366
128	Bihar	East Champaran	0.128366
776	Rajasthan	Jaipur	0.128365
233	Gujarat	Ahmedabad	0.128364
146	Bihar	Patna	0.128364
163	Bihar	West Champaran	0.128364
572	Maharashtra	Solapur	0.128363
226	Delhi	West Delhi	0.128363
941	Uttar Pradesh	Ghaziabad	0.128363
924	Uttar Pradesh	Bareilly	0.128362
560	Maharashtra	Nashik	0.128361
129	Bihar	Gaya	0.128361

TOP-20 Utilities:			
	state	district	Utilities_PDI
573	Maharashtra	Thane	0.700046
1074	West Bengal	South 24 Parganas	0.700046
564	Maharashtra	Pune	0.700045
265	Gujarat	Surat	0.700042
1059	West Bengal	Murshidabad	0.700041
221	Delhi	North West Delhi	0.700035
397	Karnataka	Bengaluru	0.700034
1080	West Bengal	Uttar Dinajpur	0.700033
1063	West Bengal	North 24 Parganas	0.700031
128	Bihar	East Champaran	0.700031
776	Rajasthan	Jaipur	0.700031
233	Gujarat	Ahmedabad	0.700029
146	Bihar	Patna	0.700029
163	Bihar	West Champaran	0.700029
572	Maharashtra	Solapur	0.700027
226	Delhi	West Delhi	0.700027
941	Uttar Pradesh	Ghaziabad	0.700027
924	Uttar Pradesh	Bareilly	0.700025
560	Maharashtra	Nashik	0.700025
129	Bihar	Gaya	0.700024

TOP-20 Jobs:			
	state	district	Jobs_PDI
573	Maharashtra	Thane	0.164355
1074	West Bengal	South 24 Parganas	0.164355
564	Maharashtra	Pune	0.164353
265	Gujarat	Surat	0.164349
1059	West Bengal	Murshidabad	0.164348
221	Delhi	North West Delhi	0.164341
397	Karnataka	Bengaluru	0.164339
1080	West Bengal	Uttar Dinajpur	0.164337
1063	West Bengal	North 24 Parganas	0.164335
128	Bihar	East Champaran	0.164335
776	Rajasthan	Jaipur	0.164334
233	Gujarat	Ahmedabad	0.164332
146	Bihar	Patna	0.164332
163	Bihar	West Champaran	0.164332
572	Maharashtra	Solapur	0.164330
226	Delhi	West Delhi	0.164330
941	Uttar Pradesh	Ghaziabad	0.164329
924	Uttar Pradesh	Bareilly	0.164327
560	Maharashtra	Nashik	0.164327
129	Bihar	Gaya	0.164326

Top National Priority Index District wise (Combined All Index/needs future Hotspots)-

TOP-20 NPI (FINAL NATIONAL PRIORITY):			
	state	district	NPI
573	Maharashtra	Thane	0.231946
1074	West Bengal	South 24 Parganas	0.231945
564	Maharashtra	Pune	0.231944
265	Gujarat	Surat	0.231941
1059	West Bengal	Murshidabad	0.231940
221	Delhi	North West Delhi	0.231936
397	Karnataka	Bengaluru	0.231935
1080	West Bengal	Uttar Dinajpur	0.231933
128	Bihar	East Champaran	0.231932
1063	West Bengal	North 24 Parganas	0.231932
776	Rajasthan	Jaipur	0.231932
146	Bihar	Patna	0.231930
163	Bihar	West Champaran	0.231930
233	Gujarat	Ahmedabad	0.231930
226	Delhi	West Delhi	0.231929
941	Uttar Pradesh	Ghaziabad	0.231928
572	Maharashtra	Solapur	0.231928
924	Uttar Pradesh	Bareilly	0.231927
129	Bihar	Gaya	0.231927
560	Maharashtra	Nashik	0.231926

State wise Data-

===== PDI =====

Top 10 States:

	state	PDI
5	Bihar	0.146351
12	Delhi	0.146351
42	Uttar Pradesh	0.146350
26	Maharashtra	0.146348
37	Rajasthan	0.146345
49	West Bengal	0.146344
25	Madhya Pradesh	0.146343
14	Gujarat	0.146342
7	Chhattisgarh	0.146342
22	Kerala	0.146342

Top 5 Districts of Each Top State:

>> Bihar – Top Districts:

	state	district	PDI
128	Bihar	East Champaran	0.146378
146	Bihar	Patna	0.146375
163	Bihar	West Champaran	0.146375
129	Bihar	Gaya	0.146369
142	Bihar	Muzaffarpur	0.146367

>> Delhi – Top Districts:

	state	district	PDI
221	Delhi	North West Delhi	0.146383
226	Delhi	West Delhi	0.146372
223	Delhi	South Delhi	0.146366
214	Delhi	East Delhi	0.146362

===== Education_PDI =====

Top 10 States:

	state	Education_PDI
5	Bihar	0.020550
42	Uttar Pradesh	0.020550
12	Delhi	0.020550
37	Rajasthan	0.020549
25	Madhya Pradesh	0.020549
28	Meghalaya	0.020549
14	Gujarat	0.020549
26	Maharashtra	0.020549
39	Tamil Nadu	0.020549
49	West Bengal	0.020549

Top 5 Districts of Each Top State:

>> Bihar – Top Districts:

	state	district	Education_PDI
129	Bihar	Gaya	0.020554
128	Bihar	East Champaran	0.020553
146	Bihar	Patna	0.020553
163	Bihar	West Champaran	0.020553
159	Bihar	Sitamarhi	0.020553

>> Uttar Pradesh – Top Districts:

	state	district	Education_PDI
918	Uttar Pradesh	Bahraich	0.020553
914	Uttar Pradesh	Azamgarh	0.020552
906	Uttar Pradesh	Agra	0.020552
995	Uttar Pradesh	Varanasi	0.020552
944	Uttar Pradesh	Gorakhpur	0.020552

===== Transport_PDI =====

Top 10 States:

	state	Transport_PDI
5	Bihar	0.128353
12	Delhi	0.128352
42	Uttar Pradesh	0.128352
26	Maharashtra	0.128351
37	Rajasthan	0.128349
49	West Bengal	0.128349
25	Madhya Pradesh	0.128348
7	Chhattisgarh	0.128348
14	Gujarat	0.128348
22	Kerala	0.128348

Top 5 Districts of Each Top State:

>> Bihar – Top Districts:

	state	district	Transport_PDI
128	Bihar	East Champaran	0.128366
146	Bihar	Patna	0.128364
163	Bihar	West Champaran	0.128364
129	Bihar	Gaya	0.128361
142	Bihar	Muzaffarpur	0.128360

>> Delhi – Top Districts:

	state	district	Transport_PDI
221	Delhi	North West Delhi	0.128369
226	Delhi	West Delhi	0.128363
223	Delhi	South Delhi	0.128360
214	Delhi	East Delhi	0.128358
225	Delhi	South West Delhi	0.128357

===== Utilities_PDI =====

Top 10 States:

	state	Utilities_PDI
5	Bihar	0.700012
12	Delhi	0.700011
42	Uttar Pradesh	0.700011
26	Maharashtra	0.700010
37	Rajasthan	0.700007
49	West Bengal	0.700007
25	Madhya Pradesh	0.700005
7	Chhattisgarh	0.700005
14	Gujarat	0.700005
22	Kerala	0.700005

Top 5 Districts of Each Top State:

>> Bihar – Top Districts:

	state	district	Utilities_PDI
128	Bihar	East Champaran	0.700031
146	Bihar	Patna	0.700029
163	Bihar	West Champaran	0.700029
129	Bihar	Gaya	0.700024
142	Bihar	Muzaffarpur	0.700023

>> Delhi – Top Districts:

	state	district	Utilities_PDI
221	Delhi	North West Delhi	0.700035
226	Delhi	West Delhi	0.700027
223	Delhi	South Delhi	0.700023
214	Delhi	East Delhi	0.700020
225	Delhi	South West Delhi	0.700018

```

===== Jobs_PDI =====

Top 10 States:
state Jobs_PDI
5 Bihar 0.164309
12 Delhi 0.164309
42 Uttar Pradesh 0.164308
26 Maharashtra 0.164306
37 Rajasthan 0.164302
49 West Bengal 0.164302
25 Madhya Pradesh 0.164300
7 Chhattisgarh 0.164300
14 Gujarat 0.164300
22 Kerala 0.164300

Top 5 Districts of Each Top State:

>> Bihar – Top Districts:
state district Jobs_PDI
128 Bihar East Champaran 0.164335
146 Bihar Patna 0.164332
163 Bihar West Champaran 0.164332
129 Bihar Gaya 0.164326
142 Bihar Muzaffarpur 0.164324

>> Delhi – Top Districts:
state district Jobs_PDI
221 Delhi North West Delhi 0.164341
226 Delhi West Delhi 0.164330
223 Delhi South Delhi 0.164324
214 Delhi East Delhi 0.164320
225 Delhi South West Delhi 0.164318

```

Top National Priority Index State Wise (Combined All Index/needs future Hotspots)-

```

===== NPI =====

Top 10 States:
state NPI
12 Delhi 0.231915
5 Bihar 0.231915
42 Uttar Pradesh 0.231914
26 Maharashtra 0.231913
49 West Bengal 0.231912
37 Rajasthan 0.231911
25 Madhya Pradesh 0.231909
14 Gujarat 0.231909
7 Chhattisgarh 0.231909
39 Tamil Nadu 0.231909

Top 5 Districts of Each Top State:

>> Delhi – Top Districts:
state district NPI
221 Delhi North West Delhi 0.231936
226 Delhi West Delhi 0.231929
223 Delhi South Delhi 0.231924
214 Delhi East Delhi 0.231922
225 Delhi South West Delhi 0.231921

>> Bihar – Top Districts:
state district NPI
128 Bihar East Champaran 0.231932
146 Bihar Patna 0.231930
163 Bihar West Champaran 0.231930
129 Bihar Gaya 0.231927
142 Bihar Muzaffarpur 0.231925

```

12. Issues faced and fixes

This project faced several challenges, but each issue was addressed to ensure smooth performance and accurate results.

Issues and Solutions:

- **Data Loading Errors:**
Some Census files had inconsistent formats. These were handled using pandas error-handling tools to skip problematic rows.
- **Normalization Problems:**
Large districts were dominating the results. This was fixed by normalizing all values by total population to create fair comparisons.
- **Chart Overlaps:**
Some charts had bars that were too close. A zoom feature was added in Matplotlib to make small differences visible.
- **Missing Libraries:**
Seaborn was not available at first. Installing it using pip solved the issue.
- **Merge Errors:**
Some districts were missing in the Aadhaar datasets. This was fixed by using outer joins and filling missing values with zero.

These fixes improved accuracy and stability throughout the project.

13. Limitations and ethical considerations

Limitations:

- Data Age:
The Census data is from 2011, so it does not reflect current conditions. Aadhaar data is more recent but still aggregated.
- Granularity:
The system uses district-level data, so very local variations at village or street level cannot be captured
- Missing Economic Data:
Metrics such as GDP or income are not included, which could provide more accurate prioritization.
- Assumptions:
The model assumes Aadhaar activity directly reflects service demand, which may vary in some cases.

Ethical Considerations:

- Privacy:
All datasets are aggregated and anonymized. No personal data is used at any point in the project.
- Fairness:
Normalization prevents larger districts from overshadowing smaller ones. The focus is on supporting underserved areas.
- Transparency:
All formulas, weights, and logic used to create the PDIs and NPI are clearly explained.

Impact:

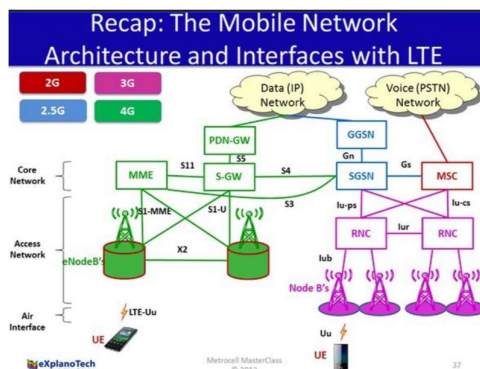
The aim is to improve development and reduce inequality by highlighting true needs.

14. Future scope

This project has several ways to grow and improve in the future.

Improvements:

- Economic Integration:
Adding GDP and income data can help adjust PDIs to reflect economic capacity and improve accuracy.
- Better Maps:
GIS tools can be used to create more detailed maps with city, road, and region-level insights.
- AI and Machine Learning:
Predictive models can forecast future demand trends and group districts with similar development patterns.
- More Detailed Data:
Working with block-level or village-level data can help in making hyper-local decisions.
- Global Expansion:
The same model can be adapted for other countries with similar identity or demographic datasets.
- Real-Time Integration:
Linking the system with live Aadhaar update feeds can help generate instant insights.



15. Conclusion

The National Population Intelligence System (NPIS) brings together Census and Aadhaar data to produce fair and useful rankings for sectors such as telecom, education, transport, utilities, and jobs. By normalizing all variables and applying logical formulas, the system provides clear and actionable insights. The results help identify key hotspots, such as urban regions needing telecom upgrades and rural regions needing more education support. The dashboard makes the data easy to understand, and the system is built with ethical practices and scalability in mind. Overall, NPIS supports India's goal of using data to drive more equal and effective development.

16. References / Appendix

References:

UIDAI Hackathon Guidelines: <https://event.data.gov.in/challenge/uidai-data-hackathon-2026>

Census of India 2011: <https://censusindia.gov.in>

Pandas Documentation: <https://pandas.pydata.org>

Matplotlib Guide: <https://matplotlib.org>

Appendix:

Code snippets from important parts of analysis.py

Sample rows from the input CSV files

Descriptions of the dashboard charts (for example, “Fig. 1: Top Districts Bar Chart”)

Complete rankings of districts and states for all PDIs and the NPI (replace with actual results)